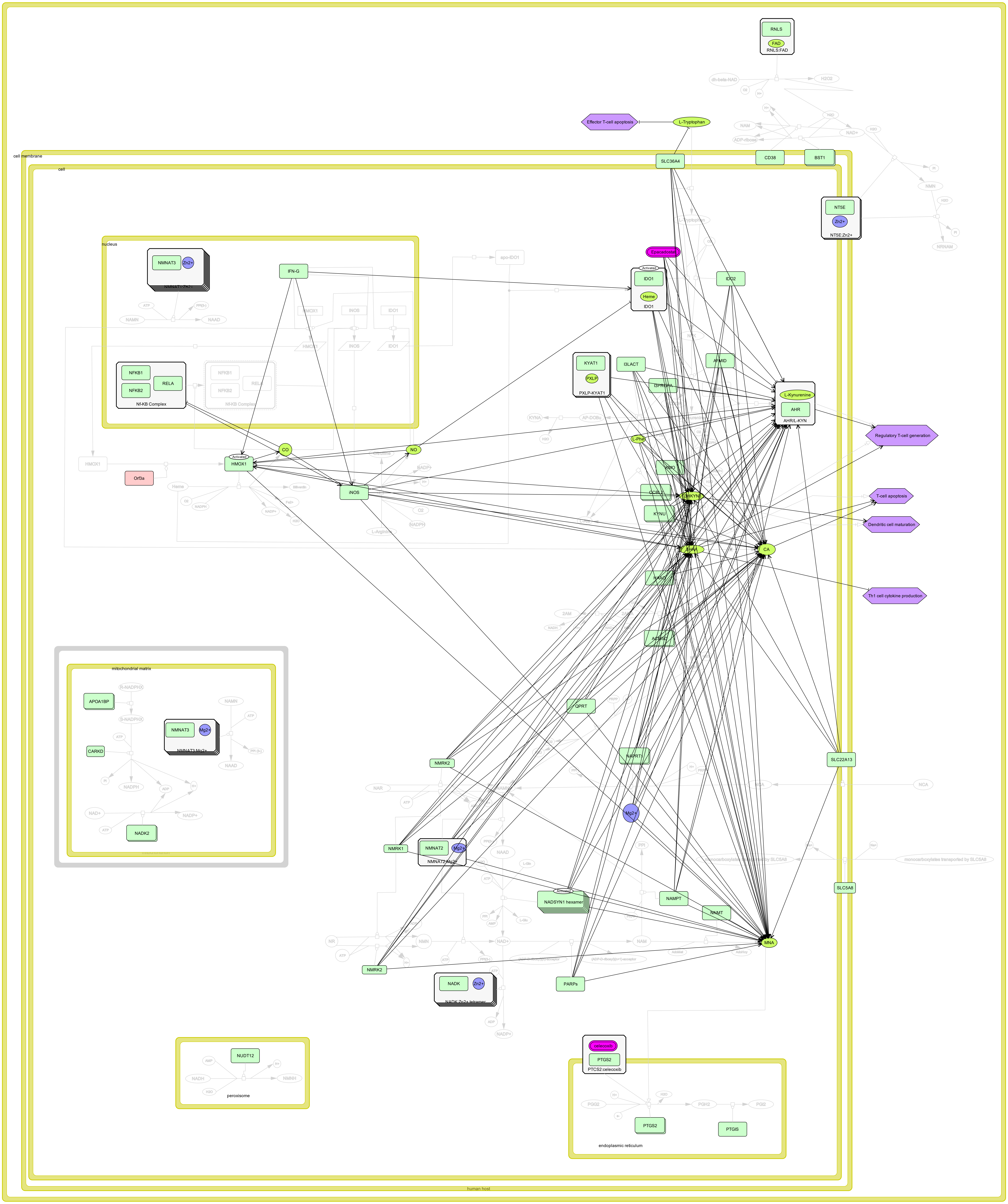
human host
cell membrane
cell
nucleus
RNLS
FAD
RNLS:FAD
dh-beta-NAD
O2
H+
H2O2
Effector T-cell apoptosis
L-Tryptophan
H+
H2O
NAM
NAD+
ADP-ribose
H2O
CD38
BST1
SLC36A4
Pi
NMN
H2O
NT5E
Zn2+
NT5E:Zn2+
Pi
NRNAM
NMNAT3
Zn2+
NMNAT1:Zn2+
ATP
PPi(3-)
NAMN
NAAD
IFN-G
HMOX1
iNOS
IDO1
HMOX1
iNOS
IDO1
apo-IDO1
Epacadostat
Activated
IDO1
Heme
IDO1
IDO2
NFKB1
NFKB2
RELA
Nf-KB Complex
NFKB1
NFKB2
RELA
Nf-KB Complex
KYAT1
PXLP
PXLP-KYAT1
I3LACT
AFMID
I3PROPA
L-Kynurenine
AHR
AHR/L-KYN
KYNA
AP-DOBu
H2O
L-Phe
Regulatory T-cell generation
HMOX1
Orf3a
Activated
HMOX1
CO
NO
Citrulline
iNOS
Heme
O2
NADPH
Biliverdin
Fe2+
NADP+
H2O
NADP+
H+
O2
NADPH
L-Arginine
KMO
CCBL2
3HKYN
KYNU
L-Ala
T-cell apoptosis
Dendritic cell maturation
3HAA
CA
HAAO
Th1 cell cytokine production
2AM
NADH
H+
2AMA
ACMSD
mitochondrial matrix
R-NADPHX
APOA1BP
S-NADPHX
NMNAT3
Mg2+
NMNAT3:Mg2+
NAMN
ATP
CARKD
ATP
PPi (3-)
NAAD
Pi
NADPH
ADP
H+
NAD+
NADP+
ATP
NADK2
mitochondrion
QPRT
PRPP
NAPRT1
SLC22A13
NMRK2
NAR
ATP
NAMN
NCA
NCA
Mg2+
PPi
NMRK1
NMNAT2
Mg2+
NMNAT2:Mg2+
NAAD
PPi
monocarboxylates transported by SLC5A8
monocarboxylates transported by SLC5A8
Na+
Na+
SLC5A8
L-Gln
ATP
Activated
NADSYN1 hexamer
NAMPT
NNMT
PPi
AMP
L-Glu
NR
NMN
NAD+
NAM
MNA
ATP
H+
ATP
PPi(3-)
(ADP-D-ribosyl)(n)-acceptor
(ADP-D-ribosyl)(n+1)-acceptor
AdoMet
AdoHcy
NMRK2
NADK
Zn2+
NADK:Zn2+ tetramer
ATP
ADP
NADP+
PARPs
NUDT12
AMP
H+
NADH
NMNH
H2O
peroxisome
celecoxib
PTGS2
PTCS2:celecoxib
H+
H2O
PGG2
e-
PGH2
PGI2
PTGS2
PTGIS
endoplasmic reticulum

PTGIS
celecoxib
PTGS2
PTCS2:celecoxib
PTGS2
NUDT12
NFKB1
NFKB2
RELA
Nf-KB Complex
NMNAT3
Zn2+
NMNAT1:Zn2+
IFN-G
SLC5A8
NADK
Zn2+
NADK:Zn2+ tetramer
BST1
NT5E
Zn2+
NT5E:Zn2+
Orf3a
CCBL2
CD38
RNLS
FAD
RNLS:FAD
APOA1BP
NMNAT3
Mg2+
NMNAT3:Mg2+
mitochondrial matrix
CARKD
NADK2
iNOS
NO
Activated
HMOX1
HAAO
NAPRT1
NMRK1
Mg2+
PARPs
QPRT
KYAT1
PXLP
PXLP-KYAT1
ACMSD
AFMID
NMNAT2
Mg2+
NMNAT2:Mg2+
I3PROPA
Activated
NADSYN1 hexamer
IDO2
NAMPT
SLC22A13
I3LACT
Activated
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Heme
IDO1
KMO
L-Phe
Epacadostat
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